African Leadership University
Kigali Innovation City, Free Economic Zone, Casabo, Kigali

Benchmarking Classical and Deep Sequential Models for Kinyarwanda News Classification on KINNEWS

Summative Project

By

Principie Cyubahiro

p.cyubahiro@alustudent.com

BSc Software Engineering

Machine Learning Techniques I

African Leadership University, Kigali

October 2026

Table of Contents

Abstract **3**
1. Introduction 4
2. Related Work **4**
3. Dataset and Data Preparation **5**
3.1 Source and structure 5
3.2 Preprocessing 6
3.3 Split strategy 6
4. Exploratory Analysis **7**
4.1 Class distribution 7
4.2 Length distribution 7
4.3 Vocabulary and distinctive words 8
5. Methodology **8**
5.1 Classical baselines 8
5.2 Deep sequential models 9
5.3 Training setup 9
6. Experiments and Results **10**
7. Error Analysis and Discussion **12**
7.1 Train versus test distribution 12
7.2 Per-class performance 12
7.3 Representative confusions 14
8. Deployment **14**
9. Limitations and Future Work **15**
10. Conclusion **16**
11. Description of Existing Resources and Implemented Work **16**
11.1 Existing resources used 16
11.2 Work implemented 17
12. References **18**
13. Project Links **19**

Abstract

The aim of this project is to build , evaluate, and deploy a Kinyarwanda news classification system given the KINNEWS corpus released by Niyongabo, Qu, Kreutzer and Huang (2020). The model is multi-class classification across 14 different categories. 7 experiments are conducted to ensure a final model deployed is robust enough to be exposed to the real world. The model development compares different algorithms including TF-IDF with Logistic Regression and Linear SVM, BiGRU and 1D CNN with three variants, self trained Word2Vec, and the author's pretrained W2V-Kin-50 and W2V-Kin-100. All experiments share the same data preprocessing pipeline and data splitting strategy of 90/10 for train and validation and the test set remains the one provided inside the test.csv file. Due to the various reasons such as class imbalance, the main metric used to evaluate the models is macro F1. The best test accuracy was 71.10% accuracy and got 57.54% for macro F1 with TF-IDF Logistic Regression where the classes were balanced. Deep sequential models achieve high validation accuracy of above 85%, but their overall validation performance is low on the test set, around 30 points lost due to a measured class-distribution mismatch in their train-valid-test split (e.g. politics is 26% of training vs 10% of test, and for economy, there is 25% of train, and 5% of test). The best model is exported and deployed with a Streamlit web app that takes in the text written in Kinyarwanda and outputs the predicted category.

Keywords: text classification, Kinyarwanda, low-resource NLP, KINNEWS, TF-IDF, BiGRU, Word2Vec, class imbalance, Streamlit.

1. Introduction

High-resource languages such as English and French have abundant text data on the web, and most NLP tools were built on top of that resource base. African languages, like Kinyarwanda, are found on the other side of that gap with less data, fewer pretrained models and tools. This project fits into this space.

The task is news classification of Kinyarwanda language: given a news article in Kinyarwanda, classify it into one of 14 categories defined as (politics, sport, economy, health, entertainment, history, technology, tourism, culture, fashion, religion, environment, education, relationship). The practical application is tangible: An article is published for example under a category section on a Rwandan news site like igihe.com, and today each article is tagged by hand before going live. This can be automated using a given article as input to the classifier to return a predicted value for the article's category.

The objective of this project is three-fold. First, develop a labelled corpus aware pipeline for preprocessing the raw KINNEWS text, then divide the document into training, validation and test sets suitable for fair testing. Second, train and compare both classical and deep sequential models, in the same experimental conditions. Third, to deploy the best model behind a small web interface that an end user will interact with. The report is organized according to the project rubric and the following parts are clearly marked: dataset preparation, methodology, results, error analysis, deployment, limitations.

2. Related Work

The direct reference to this project is Niyongabo, Qu, Kreutzer and Huang (2020) who present the corpora KINNEWS and KIRNEWS as a benchmark for low-resource monolingual and cross-lingual multiclass classification of Kinyarwanda and Kirundi

news. Their paper presents a few baselines; one using their 50-dimensional Word2Vec embeddings, which got to an accuracy of 88.65% on the KINNEWS test set, another using TF-IDF plus a Linear SVM at an accuracy of 88.53%, and a Logistic Regression at an accuracy of 87.14%.

The classic models side of this report involves Logistic Regression and the LIBLINEAR implementation of linear SVM (Fan et al., 2008) along with the TF-IDF representation from scikit-learn (Pedregosa et al., 2011). TF-IDF has been successfully proved to be superior to learned representations in multi-class text classification over short documents, particularly in low-resource scenarios, where learned representations have limited training data to generalise across.

The deep models side uses word representations trained by the Word2Vec algorithm (Mikolov, Chen, Corrado and Dean, 2013) using the package gensim (Rehurek and Sojka, 2010). Sequential modelling refers to Gated Recurrent Unit (Cho et al., 2014) encased in a bidirectional layer to provide context to the tokens on both sides. Different inductive biases including a one-dimensional convolutional neural network model is introduced, after Kim (2014) who demonstrated that Topic-classification tasks can be approached and outperformed with simple word-level CNNs against recurrent models. In Section 9, some discussion on African language models developed via transformers (AfroXLMR, Alabi et al., 2022) is observed as a direction to follow in future.

3. Dataset and Data Preparation

3.1 Source and structure

The KINNEWS dataset has been obtained from the public KINNEWS-KIRNEWS Corpus by Andrews et al., available from the authors' public GitHub repository (https://github.com/Andrews2017/KINNEWS-and-KIRNEWS-Corpus). It is a collection of 21,268 news articles in Kinyarwanda language from 15 websites and 5 newspapers in Rwanda, with supervision of 14 classes. The authors created two versions: one of them was a raw version with six columns (class index, Kinyarwanda label, English label, URL, title and content), while the other was a cleaned version, with only three columns (class index, title and content). In this

project, the raw version was employed since the implementation of pre-processed versions would obscure the control and documented pre-processing. The authors' split of 17,014 training articles and 4,254 test articles was respected: the training file was the only source for both model training and validation, and the test file was never seen until final evaluation.

3.2 Preprocessing

A missing-value check revealed no missing fields in the training file and five test rows with a missing content field. The remaining five rows of data were removed prior to further processing, reducing the test sample to 4249 articles (which is equivalent to 0.12 percent). Each article undergoes a preprocessing function, which has the following steps:

- Concatenate title and content into a single string, since the title often carries a strong category signal.
- Lowercase the full text.
- Replace URLs with a space, then replace anything that is not a lowercase letter or whitespace with a space.
- Collapse multiple spaces into one.
- Tokenise on whitespace.
- Remove tokens that appear in the 80-word Kinyarwanda stopword set released with the KINNEWS repository.
- Drop tokens shorter than two characters.
- Rejoin the surviving tokens into a cleaned string.

The same function is reused in the Streamlit application to ensure that user input to the training and inference have the same format.

3.3 Split strategy

The test set used by the authors was retained for a fair comparison of the authors' results. A stratified 10% sample of the training set was isolated as a validation set, resulting in 15,312 training articles and 1,702 validation articles, but keeping the class ratios that were found in the training set. Even in the smaller classes stratification gave a proportion in validation.

4. Exploratory Analysis

4.1 Class distribution

The class distribution of the training set is heavily imbalanced, and importantly the test split does not share the same distribution. Figure 1 shows the share of each class in training (including the carved-out validation slice) against its share in test. Politics and economy together make up roughly 51% of training articles but only 15 percent of test, while several small classes such as religion, relationship, environment, fashion and culture are substantially over-represented in test relative to training. This mismatch is the single most important dataset-level fact in this project and reappears in the error analysis in Section 7.

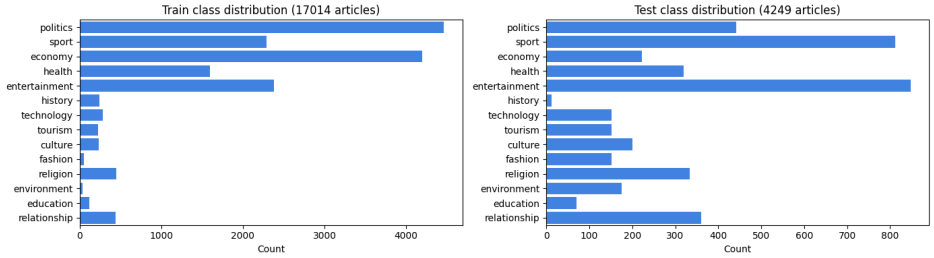


Figure 1. Class distribution in the training split (left, 17,014 articles) and the authors' held-out test split (right, 4,249 articles after dropping 5 missing-content rows). Notice how politics and economy categories dominate training but shrink in test, while classes such as entertainment, religion, relationship and environment grow in test.

4.2 Length distribution

After preprocessing, the training articles have a mean length of 213 tokens, a median of 178 tokens and a 95th percentile of 436 tokens. The test articles are measurably longer: mean 273 tokens, median 223 tokens, 95th percentile 606 tokens. Figure 2 shows the comparison. The sequence padding length was set to 650, chosen to cover the test 95th percentile with a small margin.

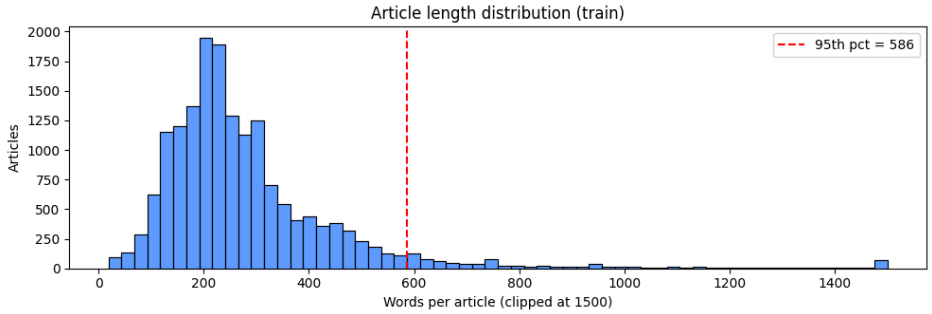


Figure 2. Article length distribution over the training set (words per article, clipped at 1500 for display). The red dashed line marks the 95th percentile at 586 words, which was used to inform the choice of sequence padding length.

4.3 Vocabulary and distinctive words

There are 150,348 distinct tokens in preprocessed training set. The vocabulary for the Keras tokenizer was set at 80,000, but the top 77,759 words account for 98% of the distribution of tokens that this vocabulary comprises. While the rate of out of vocabulary (OOV) tokens is moderate at 8.9%, it is not zero.

As part of the exploratory work, a class-level TF-IDF analysis was performed: all articles within a class were treated as one single large document and the tokens within each document were ranked according to their TF-IDF score. No single trivial token is dominant in a class, and the top distinct tokens had instances with the intuitive meaning of the class (sport for sport, religion for religion, etc), indicating that there is category signal in the preprocessed text and that a separate trivial token does not dominate a class.

5. Methodology

5.1 Classical baselines

Two classical baselines were trained on an identical TF-IDF representation of the preprocessed training text, using up to 50,000 features (unigrams and bigrams with min-doc-freq >= 2). The first baseline is L2 regularised logistic regression (C is equal to 1.0). Second one is a Linear SVM (LIBLINEAR, C = 1.0). The Logistic Regression classifier were tested twice, once using default class weights and once using class weights that were balanced to correct the imbalance. Direct

measurement of the impact of class imbalance correction was achieved and led to using class balance approach for the rest of the models.

5.2 Deep sequential models

Four deep models were trained using a 650 frame padded, tokenized sequence. The three variants applied a Bidirectional GRU consisting of 64 hidden units, the embedding matrix of which is taken from different sources:

- Experiment E4 applied a 50-dimensional Word2Vec embedding trained from scratch on the preprocessed training texts with gensim (skip-gram, 10 epochs and window 5, minimum count 2).
- For Experiment E5, the authors' pretrained embedding for words W2V-Kin-50 was used, which is released alongside KINNEWS in plain text format.
- Experiment E6 used higher dimensionality in the model with authors' pretrained embedding of W2V-Kin-100, with the same source.

The fourth deep model (Experiment E7) was obtained on top of the W2V-Kin-100 embedding, replacing the BiGRU with a 1D conv. block 128 filters, kernel size 5, global max pooling. This provided a different inductive bias since those n-gram patterns were found to be more local-level than global-level. Following the embedding, in all deep models, the embedding was followed by a SpatialDropout1D (rate 0.3) operation and before the last dense softmax with 14 output classes, there was a Dropout (rate 0.4) operation.

5.3 Training setup

All deep models were optimised with Adam using sparse categorical cross-entropy. Training lengths were up to 20 epochs using EarlyStopping on validation loss (patience: 4 with restoring best weights). Batch size was 64. For all the deep models, simple class weights were computed based on the training labels using the function compute_class_weight of scikit-learn and fed to the models to ensure balanced class weight for each class in the loss. Pretrained embedding matrices (W2V-Kin-50 and W2V-Kin-100) were left to be fine-tuned

during training (trainable = true), so they have freedom in adapting the embeddings to the sought after task.

The following evaluation parameters were used: accuracy, macro precision, macro recall, macro F1 and per-class F1 with support counts. F1 was macro averaged because it was the most important metric as many of the 14 classes were highly imbalanced and only using the accuracy metric would mask problems on small classes.

6. Experiments and Results

Table 1 lists the seven experiments with the evaluation metrics, sorted by macro F1 (the primary metric).

Experiment	Accuracy	Macro P	Macro R	Macro F1
E2 TF-IDF + LogReg (balanced)	0.7110	0.6478	0.6249	0.5754
E3 TF-IDF + Linear SVM (balanced)	0.7119	0.6395	0.5737	0.5415
E1 TF-IDF + LogReg	0.6540	0.5424	0.4596	0.4151
E6 BiGRU + W2V-Kin-100	0.5670	0.4856	0.4311	0.4086
E7 CNN + W2V-Kin-100	0.4361	0.5120	0.4657	0.4069
E5 BiGRU + W2V-Kin-50	0.5291	0.4546	0.4501	0.3971
E4 BiGRU + Own W2V-50	0.4912	0.4741	0.4458	0.3870

The figure below shows the per-class F1 of the best model.

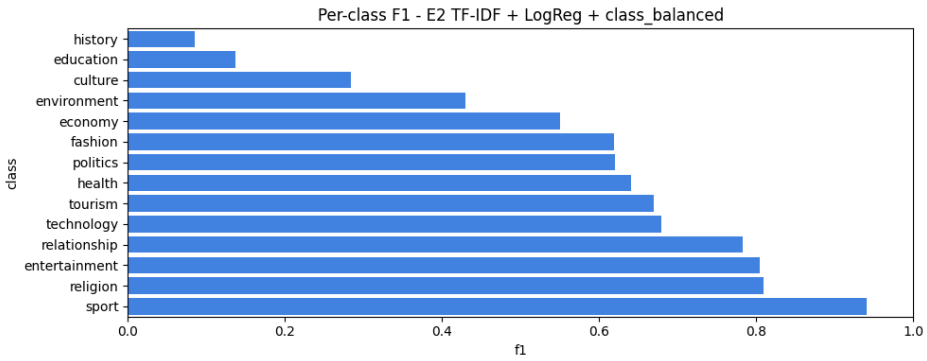


Figure 3. Per-class F1 of the best model (E2 TF-IDF + Logistic Regression with balanced class weights) on the test set, sorted from worst to best. Sport is near-perfect (F1 = 0.94) while history (12 samples) and education (70 samples) sit near the floor.

The following observations can be made based on the table and figure. The two balanced classical models (E2 and E3) rank first in terms of accuracy and macro F1. Second, there is an accuracy gain of +5.7 and a gain of +16 points of macro F1 for Logistic Regression (E2 vs E1) when the class weights are set properly, confirming that some actual work is going into the imbalance correction. Third is that no matter which model (all 4 deep sequential) you use, you will find that they all get an accuracy worse than the balanced classical baselines, despite being better than 85 percent during the training validation set. The gap will be explored in greater depth in section 7.

As expected, within the deep models the best one is BiGRU with W2V-Kin-100 (E6), which harnesses the higher dimensional larger corpus representation model than an in-domain 50 dimension Word2Vec representation that was trained on the KINNEWS training set. The 1D CNN with the same embedding (E7) is on par with the BiGRU variants in terms of macro F1, but suffers by a significant margin in terms of accuracy, which is to predict minority classes more aggressively.

7. Error Analysis and Discussion

7.1 Train versus test distribution

A diagnostic cell in the notebook evaluated the TF-IDF Logistic Regression model (trained once, no re-training) on both the validation and the test splits. For the same model, the validation accuracy was achieved as 87.78% and testing accuracy of 65.40%. Validation is a stratified 10% cut of the training set, thus it has similar distributional properties to the training set; test is the author's held out set. The difference between two splits of an apparently identical corpus of 22 points is a good indication that training and test are not merely a game of chance supplied by random noise.

It's confirmed by measuring the class distribution of each split. Politics has a 15.85-point decrease in training (-26.25%) but an increase in test (+10.40%). Economy dips 19.44 points lower for training with a drop of 24.66% to 0.52% of the test. A few small classes only go in the other direction: fashion: from 0.32% to 3.55%, environment: from 0.22% to 4.12%, religion: from 2.63% to 7.84%, relationship: from 2.59% to 8.47%, and culture: from 1.37% to 4.73%. Two large classes (third form classes) do dominate the two largest, and certain minority classes are significantly over-represented in tests. A system would mis-predict on this test set systematically because it learned the training prior.

7.2 Per-class performance

Table 2 shows per-class F1 for the best performing model (E2) and number of test articles per each class. Classes are listed in the order from worst to best F1.

Class	F1	Test support
history	0.0853	12
education	0.1373	70
culture	0.2846	201
environment	0.4298	175
economy	0.5510	222
fashion	0.6189	151

Class	F1	Test support
politics	0.6205	442
health	0.6407	320
tourism	0.6693	152
technology	0.6795	151
relationship	0.7826	360
entertainment	0.8048	848
religion	0.8099	333
sport	0.9409	812

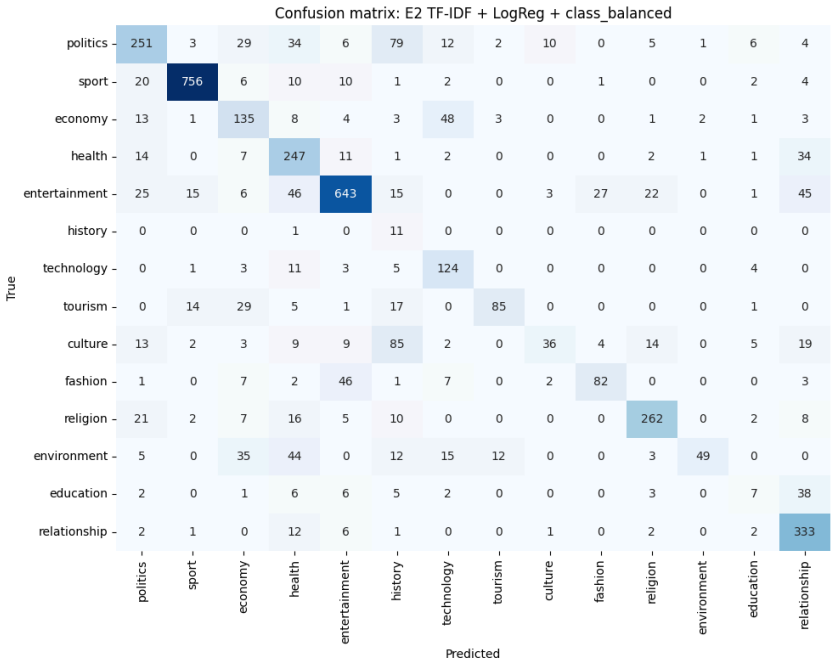


Figure 4. Confusion matrix of the best performing model (E2) on the 4,249-article test set.

Sport is almost perfectly classified (F1 of 0.9409), which matches the intuition that sport articles use a highly distinctive vocabulary (team names, scores, competition names). Both entertainment and religion score well. The worst-performing classes cluster around topics with semantic overlap: history is tiny in test (only 12 articles, F1 of 0.0853) and shares vocabulary with politics and culture; education is confused with health and relationship; culture is confused with history. Notice that the overall accuracy (71.10%) is looking healthy because

sport combined with entertainment contributes a little more than 39% of the test set, which makes overall accuracy above macro f1.

7.3 Representative confusions

Studying a sample of misclassified test pieces from the notebook revealed some common trends, for instances:

- For the articles on historical political situations, for example commemorating the Rwandan Genocide, there is shared vocabulary concerning dates, memorials and figures referring to the past, so they can be described as predicting history.
- Economy articles about government sanctions or international debt policy are predicted politics, since they mention government actors and leaders.
- Entertainment articles about a political figure making a public appearance (singing on stage during a campaign) are predicted politics, because the main named entity is a political one.
- Culture articles describing historical family dynasties are predicted history, since they describe ancestral lineage.

These confusions are not symptoms of a broken model. They are symptoms of a classification problem where the categories are not mutually exclusive at the vocabulary level and where the test prior differs from the training prior. The model is trained to make sensible choices for the training distribution but is asked to make sensible choices for the test distribution.

8. Deployment

The best model (E2 TF-IDF Logistic Regression with balanced class weights) has been exported into a deploy folder which includes the following five files: the pickled model, model.pkl; the pickled TF-IDF vectoriser, vectorizer.pkl; a label mapping file, label_map.json; a copy of the Kinyarwanda stopword set, stopwords.txt; and a small deployment config file, config.json recording the model name, model type (classical), tokenisation parameters and the number of classes in the model.

The bundle will be loaded when the Streamlit application is started and will be displayed upon initialization: a text area where user types a Kinyarwanda news article, a Classify button, and a predicted category, in addition to a bar chart showing the probability over all 14 categories on submission of the article. The function used for the pre-processing is the same function used during training, and thus the input to the application is the same as it was in the training time. The application deployed on Streamlit Community Cloud, the link to the application is given in the Project Links section at the end of this report.

9. Limitations and Future Work

The most common constraint is the difference between the train-test class distribution (measured numerically in Section 7). It places an upper bound on the going up of test scores for any model that has been trained using the authors' split of the data, and in particular it gives a tight limit for test scores of deep models whose capacity forces them to learn the training distribution more aggressively. This at the dataset level would call for either a rebalance of the test set, or augmentation of the training set with extra articles from under-represented classes.

The second is using word level tokenisation, limiting the vocabulary to 80,000 words and without next fallback. This results in a modest, but non-negligible, 8.9 % of measured OOV on the test set, which turns rare but useful words into a single <OOV> symbol. This can be reduced by using a subword tokeniser (SentencePiece or BPE).

Third, single random seeds (42) were used in all of the experiments. Reporting mean and standard deviation for multiple seeds would provide a better representation of the variance between runs particularly for the deep models where training is less predictable than the classical models.

Fourth, there was no cross validation. Since there is only one hold out (90/10 training/validation split), there is some sampling noise in the validation metrics.

Five fold cross validation would close the gap between models in the train phase of a model, but would not affect the train-test distribution gap.

10. Conclusion

This project created a fully-featured news classification system for Kinyarwanda in the KINNEWS corpus. Seven experiments were run under identical experimental conditions, measured on the authors' held-out test split, and the best test performance was achieved by a classical TF-IDF Logistic Regression with balanced class weights (71.10% accuracy, 57.54% macro F1). The deep sequential models (BiGRU and 1D CNN variants with three different word embeddings) achieved high validation accuracy, but did not outperform the classical baselines on the test set, indicating that there might be a class distribution mismatch between training and test, although this is not to fault deep architectures themselves. The best model is deployed through a web interface using Streamlit and is replicable from the linked GitHub repository below.

The big take away from the project is that it was important to have diagnostic measurement in place before incorporating additional complexity. It took just one notebook run to reveal the root cause at a single diagnostic cell that compared the validation accuracy with the test accuracy of the simple baseline. In other words, it would have been impossible to design further architecture tuning, without that measurement.

11. Description of Existing Resources and Implemented Work

11.1 Existing resources used

The project is built based on the KINNEWS dataset by (Niyongabo et al., 2020), the authors' pretrained W2V-Kin-50 and W2V-Kin-100 Embeddings along with an

80-word list of stop words in Kinyarwanda. The Python libraries used include scikit-learn for TF-IDF, Logistic Regression, and Linear SVM, gensim for self-trained Word2Vec, Tensorflow Keras for the deep sequential models, and Streamlit for deployment. Each resource is cited in Section 12.

11.2 Work implemented

The following project components were implemented from scratch: the full preprocessing pipeline, a stratified train/validation/test split strategy, a choice of hyperparameters (MAX_VOCAB = 80000, MAX_LEN = 650, trainable embeddings, EarlyStopping patience of 4, balanced class weights), shared helpers for the experiments, cell that exposed the test-train set mismatch, error analysis logic, the deployment bundle export, and streamlit app creation.

12. References

Alabi, J. O., Adelani, D. I., Mosbach, M., & Klakow, D. (2022). Adapting pre-trained language models to African languages via multilingual adaptive fine-tuning. In Proceedings of the 29th International Conference on Computational Linguistics (COLING 2022), 4336-4349.

Cho, K., van Merrienboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder-decoder for statistical machine translation. In Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), 1724-1734.

Fan, R.-E., Chang, K.-W., Hsieh, C.-J., Wang, X.-R., & Lin, C.-J. (2008). LIBLINEAR: A library for large linear classification. Journal of Machine Learning Research, 9, 1871-1874.

Kim, Y. (2014). Convolutional neural networks for sentence classification. In Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), 1746-1751.

Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. arXiv preprint arXiv:1301.3781.

Niyongabo, R. A., Qu, H., Kreutzer, J., & Huang, L. (2020). KINNEWS and KIRNEWS: Benchmarking cross-lingual text classification for Kinyarwanda and Kirundi. In Proceedings of the 28th International Conference on Computational Linguistics (COLING 2020), 5507-5521.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

Rehurek, R., & Sojka, P. (2010). Software framework for topic modelling with large corpora. In Proceedings of the LREC 2010 Workshop on New Challenges for NLP Frameworks, 45-50.

Streamlit Inc. (2024). Streamlit: The fastest way to build and share data apps. Software documentation, https://docs.streamlit.io.

TensorFlow Developers. (2024). TensorFlow (version 2.17). Zenodo. https://doi.org/10.5281/zenodo.4724125.

13. Project Links

- GitHub repository (source code, README, notebook, Streamlit app): https://github.com/PrincipieCyupe/Kinyarwanda-News-Classification-Model
- Deployed Streamlit application: https://kinnews.streamlit.app
- Demo video: Demo Video - MLT1 Summative Project.mp4

Notice: *When the deployed application is offline at the graduation time because of Streamlit Community Cloud quotas, the instructions about how to run it locally is provided in the README file of the GitHub repository.*